

Quantitative Trading Strategy Development Task

ML Engineer + Quantitative Researcher (Fresher Level)

Assignment Date: 14 January 2025

Submission Deadline: 18 January 2025, 11:59 PM IST

Duration: 5 Days

Objective

Build a complete quantitative trading system demonstrating your ability in data engineering, feature engineering, regime detection, algorithmic trading strategy implementation, machine learning, and statistical analysis.

Part 1: Data Acquisition and Engineering

Task 1.1: Data Fetching

Fetch 5-minute interval data for the last 1 year:

1. NIFTY 50 Spot

- OHLCV data

2. NIFTY Futures

- Current month contract
- Handle monthly expiry rollover
- Fields: OHLCV, Open Interest

3. NIFTY Options Chain

- Strikes: ATM, ATM $\pm$ 1, ATM $\pm$ 2
- Both Call and Put
- Fields: LTP, IV, Open Interest, Volume

API Options: Zerodha Kite Connect, ICICI Breeze, Groww, or NSE Historical Data

Deliverables:

- nifty_spot_5min.csv
- nifty_futures_5min.csv
- nifty_options_5min.csv

Task 1.2: Data Cleaning

Clean the fetched data:

- Handle missing values
- Remove outliers
- Align timestamps across all datasets
- Handle futures contract rollovers
- Calculate ATM strike dynamically

Deliverables:

- Cleaned datasets
- data_cleaning_report.txt

Task 1.3: Data Merging

Merge spot, futures, and options data on timestamp.

Deliverable:

- nifty_merged_5min.csv

Part 2: Feature Engineering

Task 2.1: EMA Indicators

Create the following moving averages:

- EMA (5-period) - Fast moving average
- EMA (15-period) - Slow moving average

These will be used exclusively for trading signals.

Task 2.2: Options Greeks and IV

Calculate the following Greeks for ATM Call and Put using `py_vollib` or `mibian`:

- Delta
- Gamma
- Theta
- Vega
- Rho

Use Black-Scholes model with risk-free rate of 6.5%.

Task 2.3: Derived Features

Create the following derived features:

- Average IV = $(\text{call_iv} + \text{put_iv}) / 2$
- IV Spread = $\text{call_iv} - \text{put_iv}$
- PCR (OI-based) = $\text{total_put_oi} / \text{total_call_oi}$
- PCR (Volume-based) = $\text{total_put_volume} / \text{total_call_volume}$
- Futures Basis = $(\text{futures_close} - \text{spot_close}) / \text{spot_close}$
- Returns (spot and futures)
- Delta Neutral Ratio = $\text{abs}(\text{call_delta}) / \text{abs}(\text{put_delta})$
- Gamma Exposure = $\text{spot_close} \times \text{gamma} \times \text{open_interest}$

Task 2.4: Final Feature Set

Create final dataset with all EMA, Greeks, IV, and derived features.

Deliverable:

- nifty_features_5min.csv

Part 3: Regime Detection

Task 3.1: HMM Implementation

Implement Hidden Markov Model to classify market into 3 regimes:

- Regime +1: Uptrend
- Regime -1: Downtrend
- Regime 0: Sideways

Input Features for HMM (Options-based only):

- Average IV
- IV Spread
- PCR (OI-based)
- ATM Delta
- ATM Gamma
- ATM Vega
- Futures Basis
- Spot Returns

Use hmmlearn library with 3 states. Train on first 70% of data.

Task 3.2: Regime Visualization

Create visualizations:

- Price chart with regime overlay (color-coded)
 - Transition matrix heatmap
 - Regime statistics (IV, Greeks distribution per regime)
 - Duration histogram
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Part 4: Trading Strategy

Task 4.1: Strategy Implementation

Implement 5/15 EMA strategy with regime filter:

LONG Entry:

- 5 EMA crosses above 15 EMA
- Current regime = +1
- Enter at next candle open

LONG Exit:

- 5 EMA crosses below 15 EMA
- Exit at next candle open

SHORT Entry:

- 5 EMA crosses below 15 EMA
- Current regime = -1
- Enter at next candle open

SHORT Exit:

- 5 EMA crosses above 15 EMA
- Exit at next candle open

No trades in Regime 0.

Task 4.2: Backtesting

Backtest the strategy:

- Training: First 70% of data
- Testing: Last 30% of data

Calculate metrics: Total Return, Sharpe Ratio, Sortino Ratio, Calmar Ratio, Max Draw-down, Win Rate, Profit Factor, Average Trade Duration, Total Trades.

Part 5: Machine Learning Enhancement

Task 5.1: Problem Definition

Build binary classifier to predict if a trade signal will be profitable.

Target: 1 if trade is profitable, 0 otherwise.

Features: All engineered features, regime, time-based features, lag features, signal strength.

Task 5.2: Model Training

Train two models:

Model A: XGBoost

- Use time-series cross-validation
- Train gradient boosting classifier

Model B: LSTM

- Input: Sequence of last 10 candles
- Architecture: LSTM → Dropout → Dense → Output

Task 5.3: ML-Enhanced Backtest

Backtest strategy with ML filter: Only take trades when ML model predicts profitable (confidence > 0.5).

Compare performance: Baseline vs XGBoost vs LSTM.

Part 6: High-Performance Trade Analysis

Task 6.1: Outlier Detection

Identify profitable trades beyond 3-sigma (Z-score > 3).

Analyze features: regime, IV, ATR, time of day, Greeks, trade duration, EMA gap, PCR.

Task 6.2: Pattern Recognition

Compare outlier trades vs normal profitable trades using statistical tests.

Create visualizations:

- Scatter plot (PnL vs duration)
- Box plots (feature distributions)
- Correlation heatmap
- Time distribution

Task 6.3: Insights Summary

Answer:

- What percentage are outliers?
 - Average PnL comparison
 - Regime patterns
 - Time-of-day patterns
 - IV characteristics
 - Distinguishing features
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Deliverables

1. GitHub Repository

Deploy complete project on GitHub with:

Repository Structure:

- data/ (all CSV files)
- notebooks/ (7 Jupyter notebooks: data acquisition, cleaning, feature engineering, regime detection, baseline strategy, ML models, outlier analysis)
- src/ (Python modules: data_utils.py, features.py, greeks.py, regime.py, strategy.py, back-test.py, ml_models.py)
- models/ (saved models)
- results/ (all result files)
- plots/ (all visualizations)
- requirements.txt
- README.md

README.md must include:

- Project overview
- Installation instructions
- How to run
- Project structure explanation
- Key results summary

2. PowerPoint Presentation

File: YourName_Presentation.pptx (25-30 slides)

Required Sections:

- 1. Title Slide
- 2. Executive Summary (2-3 slides)
- 3. Data Pipeline (2-3 slides)
- 4. Feature Engineering (3-4 slides)
- 5. Regime Detection (3-4 slides)
- 6. Baseline Strategy Results (3-4 slides)
- 7. ML Enhancement (4-5 slides)
- 8. High-Performance Trade Analysis (3-4 slides)
- 9. Conclusion and Recommendations (2 slides)

Professional design with clear charts, consistent colors, slide numbers.

Evaluation Criteria

Component	Evaluation Focus	Points
Data Pipeline	Data acquisition, cleaning, handling of missing values, futures rollover, merging accuracy, documentation	15
Feature Engineering	Technical indicators, Greeks calculation accuracy, IV computation, derived features creativity	20
Regime Detection	HMM implementation, regime interpretation, transition analysis, visualization quality	15
Baseline Strategy	Strategy logic implementation, backtest methodology, risk management, performance metrics	15
ML Models	Model training, validation approach, performance comparison, feature importance analysis	20
Outlier Analysis	Statistical methodology, pattern recognition, insights extraction, visualization clarity	10
Presentation	Professional design, narrative structure, clarity of communication, documentation quality	5
Total		100

Submission Instructions

Email to: Info@klypto.app

Subject: Quant Task Submission - [Your Full Name]

Submit:

1. GitHub repository link (public)
2. PowerPoint presentation (YourName_Presentation.pptx)
3. Data files (include in repo, else Google Drive link)

Email Body:

- Name, Email, Phone, Submission Date
- GitHub link
- Brief summary (2-3 sentences)

Deadline: 18 January 2025, 11:59 PM IST

Late submissions not accepted.

Important Notes

- If shortlisted, you will explain your work in interview

Good Luck!